

Shashi Bhushan Vijay

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PROFESSIONAL SUMMARY

Results-driven Integrated M.Tech Computer Science student and patent holder with hands-on experience building production-grade software systems. Proven ability to develop, test, and deploy applications across the full stack—from edge hardware integration to LLM-powered SaaS platforms. Adept at following design and code standards, writing unit tests, maintaining system documentation, and contributing to continuous improvement. Experienced in Agile and iterative development methodologies with a strong focus on risk awareness and quality delivery.

EXPERIENCE & PROJECTS

FLEX-DCA AI Platform — AI-Powered Debt Collection Allocation & Compliance System

Jan 2026 – Feb 2026

Repo: github.com/shashi-bhushan-27/fedX_hackathon

- Built a full-stack debt recovery platform using FastAPI, React, and SQLite to ingest cases, run predictions, auto-allocate DCAs, and track operational KPIs via REST APIs.
- Developed and integrated 3 CatBoost models (recovery probability, aging risk, closure speed) with financial feature engineering for real-time decision support.
- Implemented a RAG-based legal copilot using FAISS + SentenceTransformers + Groq LLaMA 3.1, enabling policy-grounded answers for FDCA, Regulation F, and RBI compliance queries.
- Created an operations dashboard with KPI monitoring and case workflows, improving visibility into recovered/open cases and allocation outcomes.
- Structured backend into modular services (ML, allocation, RAG, dashboard, ingestion) to support scalable, production-style architecture for enterprise debt operations.

AutoML Assistant – SaaS Platform

Jan 2026

Repo: github.com/shashi-bhushan-27/AutoML_Assistant

- Built a full-stack SaaS application (Streamlit + FastAPI) with well-defined component boundaries; contributed to design of individual modules including preprocessing pipeline, model training engine, and inference endpoints.
- Integrated LangChain + Groq (Llama 3.1) RAG pipeline meeting non-functional requirements for dynamic model strategy recommendations with low latency.
- Implemented a SHAP-powered explainability engine and live JSON/CSV batch inference endpoints; documented API contracts and maintained technical specifications for subject matter expert review.
- Identified and escalated a model performance risk during evaluation, aligning risk monitoring into day-to-day development workflow.

Patent: Coordinated Indoor Position Determination via Multi-Stage Signal Processing

Dec 2024 – Dec 2025

VIT, Vellore, IN | Application No. 202541115892

- Engineered a FastAPI + MQTT backend with well-defined interfaces for processing real-time signals from ESP32/Raspberry Pi edge devices; wrote unit tests validating signal processing accuracy against defined benchmarks.
- Developed a dynamic stacking ML ensemble (Scikit-learn, XGBoost) achieving 1.6 m accuracy ($R^2 = 0.978$), exceeding industry benchmarks; followed coding standards throughout and contributed design improvements via technical review sessions.
- Deployed A* graph-based navigation for optimal path generation across unseen layouts; created and maintained end-to-end system documentation supporting the patent filing process.
- Provided production-level support during testing phases, monitoring for anomalies and mitigating hardware integration risks in real time with appropriate escalation of blockers.

EDUCATION

Vellore Institute of Technology, Vellore

Integrated M.Tech in Computer Science Engineering | GPA: 7.0/10

Expected April 2027

SM Public School, Haridwar

Class XII: 91.2% | Class X: 87%

2018–21

TECHNICAL SKILLS

Languages & Core: C++, Java, Python, JavaScript, SQL

Backend & Systems: FastAPI, REST APIs, Microservices, MQTT

Testing & Quality: Unit testing, SHAP explainability, model benchmarking, endpoint validation

Software Development Methodologies: Agile, iterative development, CI/CD principles, continuous improvement

Databases & Cloud: PostgreSQL, SQLite, AWS, Git

ML & AI: Scikit-learn, XGBoost, LSTM, SHAP, RAG, FAISS, LangChain, Groq, Quantitative Finance

LEADERSHIP & EXTRACURRICULARS

Sponsorship Coordinator, ISTE Chapter – VIT Vellore

- Secured Rs. 1,00,000+ in corporate sponsorships by pitching to 100+ founders and startups; orchestrated end-to-end event logistics for flagship hackathon (Horizon).
- Developed outreach frameworks that established strategic industry connections, demonstrating strong stakeholder collaboration skills transferable to working with subject matter experts.

ACHIEVEMENTS & CERTIFICATIONS

- 1st Place, ARC Hackathon (ARC Lab, VIT) – Built a GUI-based AutoML tool that automatically benchmarks and recommends the best ML model for any given dataset. ([link](#))
- Certification: Stock Market Basics & Technical Analysis ([link](#))
- Certification: Designing Seamless Notification Service ([link](#))